Ross Jones

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EDUCATION

Queen's University
Kingston, ON
Sep 2022 - May 2026

Engineering Physics, Mechanical, B.A.Sc.

Engineering Physics, Mechanical Engineering subdiscipline.

Awards: Dean's List, PEO Scholarship, Principals Scholarship, Thames Valley Education Award.

Relevant Coursework: Thermodynamics · Fluid Mechanics · Heat Transfer · Automatic Controls · Mechatronics Engineering · Electronics · Cryogenic Engineering · Quantum Mechanics

EXPERIENCE

Hatch, *Mechanical EIT*
Mississauga, ON
Jul 2026 - Present

- Part of the Project Delivery team supporting a large-scale mineral processing facility project.
- Collaborating with a multidisciplinary team to develop and deliver mechanical design packages.
- Preparing and maintaining mechanical equipment lists, material take-offs, and project documentation.

Octane Medical Group, *Engineering Intern*
Kingston, ON
May 2025 - Aug 2025

- Developed a LabVIEW control and monitoring application for a pharmaceutical manufacturing system, integrating real-time data acquisition, hardware control, and data compliance logging.
- Optimized sterile reservoir bag filling using compressed air; conducted SOLIDWORKS flow/particle simulations to improve fluid-cell separation; performed stress analysis on cell culture flasks.

Octane Medical Group, *Research Assistant*
Kingston, ON
Apr 2024 - Aug 2024

- Contributed to the development of orthopedic implants from live tissue using tissue engineering.
- Optimized a tangential flow filtration system to concentrate cell solution for a medical device.

Smith Engineering, *Project Manager*
Kingston, ON
Aug 2024 – May 2026

- Manage 4+ student design teams (5–6 members each) in APSC 101 & 103 courses, providing technical guidance, feedback, and assessment.
- Mentor students through project definition, prototyping, testing, and documentation to ensure successful outcomes.

RESEARCH & DESIGN PROJECTS

- **Thesis:** Designed a low-cost, high-precision laser wavemeter based on Fizeau interferometry. Reflected and refracted a laser beam into a CMOS camera for fringe capture and performed computational wavelength calibration using Python.
- **Hybrid Vertical Axis Wind Turbine:** Developed a hybrid helical Darrieus–Savonius turbine to harvest wind generated by vehicles on highways. Designed and simulated the turbine in SOLIDWORKS and COMSOL to evaluate aerodynamic efficiency and optimize for kilowatt-scale power generation.
- **Solar Cell Performance Evaluator:** Designed and 3D-printed enclosure for a motorized goniometer in SOLIDWORKS; developed LabVIEW scripts for I–V tracing and photodiode characterization. Interfaced with an Arduino.
- **Gouy Balance Apparatus:** Designed an apparatus to measure material magnetic susceptibility; integrated experimental setup with analytical modeling of magnetic response.
- **EV Battery Enclosure:** Designed thermally regulated and impact-protected battery enclosure for an electric vehicle.

ADDITIONAL

Technical Skills: SOLIDWORKS, LabVIEW, AutoCAD, Python, C, MATLAB, Arduino, Microsoft Office, LTspice, Fusion 360

Design Team: Queen's Rocket Engineering Team - Worked on the aero-structures sub-team, assisting with material layups and simulations for rockets designed to compete in the Spaceport America Cup in New Mexico each spring.

Volunteer: O-Week Leader (Queen's U) Mentored 30 new engineering students. Youth Council Member (HEALYAC, Western U).

Interests: Travelling, Tennis, Running and Hiking.

References Available upon Request.